

Deterministic geometric Savitzky-Golay dynamics estimator

Shared parameter update: reference -> estimated

mass [kg]	2.351597591 -> 2.365684669	delta=+0.0140871
ratio=1.0059904		
CoG x [m]	-0.002024708562 -> -0.002836224674	delta=-0.000811516
ratio=-		
CoG y [m]	-3.052657894e-05 -> 0.0008519464097	delta=+0.000882473
ratio=-		
CoG z [m]	0.009509749599 -> 0.01088631769	delta=+0.00137657
ratio=-		
rotor effectiveness 1	1 -> 0.962812813	delta=-0.0371872 ratio=0.96281281
rotor effectiveness 2	1 -> 0.9749949283	delta=-0.0250051 ratio=0.97499493
rotor effectiveness 3	1 -> 0.9619412381	delta=-0.0380588 ratio=0.96194124
rotor effectiveness 4	1 -> 0.9516138499	delta=-0.0483862 ratio=0.95161385
rotor command lag [s]	0.004997730255 -> 0.2	delta=+0.195002 ratio=-

Inertia matrix [kg m^2]

reference:

```
[ 0.06500006148, -7.27899253e-07, 1.901508003e-05]
[-7.27899253e-07, 0.06495265634, 5.9167305e-08]
[ 1.901508003e-05, 5.9167305e-08, 0.1289921107]
```

estimated:

```
[ 0.06501360361, -1.777114535e-07, 2.400969495e-05]
[-1.777114535e-07, 0.06529953182, 1.097642959e-05]
[ 2.400969495e-05, 1.097642959e-05, 0.128994522]
```

Principal moments: reference -> estimated

principal moment 1 [kg m^2]	0.06495264516 -> 0.06501359449	delta=+6.09493e-05
ratio=1.0009384		
principal moment 2 [kg m^2]	0.06500006701 -> 0.06529953005	delta=+0.000299463
ratio=1.0046071		
principal moment 3 [kg m^2]	0.1289921163 -> 0.1289945329	delta=+2.41655e-06
ratio=1.0000187		

Optimization

joint dynamics loss: 1.15746308566
Gaussian-prior cost: 0.348655029213
selected strict-ZOH rotor lag [s]: 0.2

Bag failure_1_single

reconstruction position RMSE [m]: 0.00455405797261
reconstruction position component RMSE [m]: [0.003382 0.001131 0.002832]
reconstruction orientation RMSE [deg]: 1.63617908637
gyro consistency RMSE xyz [rad/s]: [0.074706 0.140858 0.026014]
specific-force consistency RMSE xyz [m/s^2]: [0.819556 0.149812 0.265803]
inferred external force RMS xyz [N]: [1.096201 0.392979 3.003827]
inferred external torque RMS xyz [N m]: [0.075803 0.254293 0.0884]
bag dynamics loss: 1.15746308566
gimbal command lag [s] 0.004998683929 -> 0.03226901675

Recorded command timestamp intervals (data-derived)

Bag failure_1_single

rotor: median dt=0.00499773026 s (200.091 Hz), min=2.00271606e-05 s, max=0.0138893127 s
gimbal: median dt=0.00499868393 s (200.053 Hz), min=4.529953e-06 s, max=2.48282003 s

Selected optimizer diagnostics

status=1 success=True nfev=4 njev=4
message=`gtol` termination condition is satisfied.
reported optimality=8.29991506634542e-08
initial cost=1.506118527187727
final cost=1.5061181148746856
cost reduction=4.1231304148681147e-07
coordinate step L2=0.00026337342771118235
final gradient inf-norm=8.29991506634542e-08
gradient tolerance satisfied=True
final Jacobian rank=14
final Jacobian cond(nonzero)=390.70662516062816
max directional finite-difference Jacobian relative error=7.599270019489056e-12